Verigate PortalDot

Authorization middleware for AI-agent payments on PortalDot V3

A proposal is not authority.

Verigate evaluates the action. The bridge verifies signed evidence. A dedicated Solidity guard enforces the exact approved transfer.

VERIGATE	BRIDGE	PORTALDOT
Ed25519 decision	Exact action binding	EIP-712 guard

TESTNET

420420777

PortalDot V3 EVM testnet

V2 transfer recorded • receipt/event matched

Track fit

Modular Cloud Solutions and Middleware

Built around a narrow, reusable control point between AI intent and token execution.

Preventing unrestricted agent spending

The authorization must bind the decision to the exact thing the executor is allowed to do.

01 RECIPIENT SUBSTITUTION

An approved payment must not be reused with a different destination.

02 AMOUNT TAMPERING

The amount must remain the same from signed intent to contract call.

03 REPLAY OR EXPIRY

A valid authorization should not be usable again or after its deadline.

Design principle

A signed ALLOW is necessary but not enough: its authority must be short-lived, narrowly scoped, and bound to a recipient, amount, chain and contract.

A narrow path from decision to execution

Off-chain verification and on-chain enforcement have separate, explicit trust boundaries.

1

AGENT INTENT

Proposed action

2

VERIGATE

Signed ALLOW + execution authorization

3

BRIDGE

Verify Ed25519; bind exact action

4

EIP-712

Chain, guard, recipient, amount, nonce, deadline

5

GUARD

Reject invalid/replayed or altered transfer

WARN / BLOCK / invalid signature / mismatched action

The bridge refuses to sign; the contract cannot independently override this off-chain decision.

Why PortalDot V3 — and where this project fits

Two 0.1 POT testnet transfer records exist, with different evidence availability for V1 and V2.

- **Recorded testnet transfers (V1 + V2)**

  V1: 0.1 EVM POT in block 132475; saved success receipt and matching TransferExecuted event. A later query to the same configured RPC no longer returned that historical receipt. V2: 0.1 EVM POT in block 149241; receipt, event, sender debit and recipient credit were re-queried through that same RPC.

- **EVM-compatible contract execution**

  The guard uses EIP-712 signatures bound to chain ID and contract address. V2 is the current execution demo; V1 remains a separate predecessor.

- **Substrate-aware operational checks**

  Tooling checks the configured Substrate WebSocket endpoint and genesis hash before sensitive network operations. This is not proof of EVM-to-SS58 account mapping.

TRACK

Modular Cloud Solutions and Middleware

Not claiming:

A cross-chain asset bridge, verified wallet mapping, a second independent EVM RPC provider, production readiness, or an independent audit.

V2 execution evidence on PortalDot V3

The recorded positive path is real; limits and wallet integration are not overclaimed.

NETWORK

PortalDot V3 EVM testnet

CHAIN ID

420420777

CURRENT CONTRACT

PortalDotAgentGuardV2

CONTRACT ADDRESS

0x87C494a5...10E046C6

DEPLOYMENT TX (RECORDED)

0x331c1fc0...3f5d3494b

TRANSFER TX

0xaf3df785...f2e90f5551e

TRANSFER RESULT

Success • block 149241

READ-BACK

V2 policy

Recipient allowlist

0.1 POT / transfer

0.5 POT per resettable 3600-second interval.

TRUST BOUNDARY

Ed25519 stays off-chain; V2 checks the bridge EIP-712 signature.

Evidence and unresolved limits

The V2 positive execution path is recorded; negative policy cases are local tests.

LOCAL

contract tests

ALLOW

signed artifacts accepted

BLOCK

denied path refused

- **V1 transfer evidence is historical**

  A 0.1 POT transfer was recorded in block 132475 with a successful receipt and matching TransferExecuted event. A later lookup through the same configured RPC returned no historical receipt.

- **V2 transfer re-queried**

  Receipt status 1, one matching TransferExecuted event, sender debit and 0.1 POT recipient credit were checked through the same configured RPC in block 149241.

IMPORTANT DISTINCTION

The transfer was 0.1 POT. Sender balance fell from 416.1965 to 408.3687 POT across an isolated block, exactly matching the 7.8278 POT receipt-derived amount. One configured RPC only; recipient balance increased from 0 to 0.1 POT across the isolated block. Runtime execution-fee vs refundable storage-deposit components were not separately determined.

Remaining technical and adoption gaps

The public positive path is recorded; cost semantics, live negative cases and ecosystem demand remain open.

- **Explain the sender debit**

  The sender debit reconciles to 7.8278 POT for a 0.1 POT recipient credit. This is a gross balance debit; runtime execution-fee versus refundable storage-deposit components remain unresolved.

- **Prove negative policy paths**

  The cap and allowlist rejection cases are backed by local Solidity tests, not recorded public-testnet reverts.

- **Validate ecosystem need**

  No PortalDot Wallet/DaaS integration, team endorsement, or external user adoption is currently confirmed. The current contribution is a reference middleware path; independent user demand remains to be demonstrated.

ECOSYSTEM VALUE

A reference middleware path from Verigate Ed25519 authorization off-chain to EIP-712 enforcement by a PortalDot V3 guard. This is not an integration into PortalDot Wallet or DaaS.

LIMITATIONS

Testnet prototype, not audited or mainnet. One configured RPC only; no second-provider check or EVM genesis pin. V2 cap/allowlist failures have local-test evidence, not public-chain revert transactions. The cap is resettable, not sliding-window.